Lab worksheet 6: Introduction

Instructions

1. Create a Java project in IntelliJ within your folder and name it using your student number in the following formats: **"OOP_CS_2022_XXX" or "OOP_ET_2022_XXX"**. (Eg: OOP_CS_2022_001, OOP_ET_20222_007)
2. Create distinct packages for each lab worksheet, naming them in the following format "**LW_XX**." (Eg: LW_01)
3. Create distinct classes for each question, naming them "**QX**." (Eg: Q1)
4. Upload your project files to your GitHub repository.

Questions

1. Define the Dog and the Cat classes as subclasses of the following Pet class. The speak method returns an empty string.

pet.java

```
public class Pet {
private String name;
public String getName( ) {
return name;
}


public void setName(String petName) {
name = petName;
}


public String speak( ) {
return "I'm your cuddly little pet.";
}
}
```

2. Write a Java program that creates an Array of pets. An item in the array is either a Dog or a Cat. For each pet, enter its name and type (‘c’ for cat and ‘d’ for dog). Stop the input when the string STOP is entered for the name. After the input is complete, output the name and type for each pet in the list.

3. Repeat the above question, but this time group the output by printing out the names of all cats first and then the names of all dogs.

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4. Modify the Dog class to include a new instance variable weight (double) and the Cat class to include a new instance variable coatColor (string). Add the corresponding accessors and mutators for the new instance variables. Modify the above question by inputting additional information appropriate for the type. First, you input the name and type, as before. If the type is a cat, then input its coat color. If the type is a dog, then input its weight. After the input is complete, output the name, type, and coat color for the cats and the name, type, and weight for the dogs.

5. Suppose you have an array of Dog and Cat objects from the above question and want to find the average, minimum, and maximum weight of dogs. To compute these values, you must scan the whole array. It would be more efficient if you could get the results by traversing only Dog objects in the array. One approach to achieve this improvement is to create another array that includes only Dog objects (actually references to Dog objects).create the additional dog array. Then find the average, minimum, and maximum weights of the dogs by traversing the dog array.

6. Perform the input routine for the Dog and Cat information as specified in the above question. In addition to creating a main array, create separate cat and dog arrays. After the arrays are created, allow the end user to add or remove information. Display the following menu choices:
   1. Add Cat
   2. Add Dog
   3. Remove Cat
   4. Remove Dog
   0. Quit

   Allow the user to remove the information by specifying the name (assume there are no duplicate names). When adding a new cat or a dog, input the corresponding data values. Make sure to update the arrays accordingly when adding or removing a pet. Repeat the operation until the user wants to quit. Notice that we use 0 for the menu choice Quit. By doing this way we don’t have to change the code for quitting the program when we add more menu choices later.